

## Index

### **a**

absolute inverse-predicted concentrations  
120, 121  
absorbance 6, 18, 25, 161, 240, 278, 281  
active principal ingredient (API) 271  
Akaike information coefficient (AIC)  
132  
analysis of variance (ANOVA) 13, 65–72,  
98, 187  
analytical validation 55, 227  
ANOVA classic algorithm 72–75  
a posteriori 136, 192  
authentic analyte in surrogate matrix  
12–13  
Avogadro's constant 152

### **b**

balanced and unbalanced experimental  
design 71–72, 125  
Beer's law 6, 25  
 $\beta$ -expectation tolerance interval ( $\beta$ -ETI)  
114, 118, 119, 124–128, 237, 257,  
263, 302  
 $\beta$ - $\gamma$  content tolerance interval ( $\beta$ - $\gamma$ -CTI)  
125, 128  
between-laboratories variance 61, 69  
between-series variance 61, 67, 69, 72,  
136–138, 279, 281, 296  
bias and recovery yield 82–83  
Bioanalytical Method Validation 142

### **c**

calibration  
direct and inverse 25–28  
least-squares regression method  
28–34  
metrological approach 51–53  
misuses of regression 45–48  
modes 9–10  
nonlinear calibration curve 41–44  
ordinary least-squares (OLS) regression  
34–37  
quantification and 2–3  
standard addition method (SAM)  
48–51  
weighted least-squares (WLS)  
regression 37–40  
calibrators 3–5, 12, 14, 16, 26–30, 32,  
33, 36–38, 43, 52–56, 153, 182,  
265  
capillary zone electrophoresis (CZE) 273  
certified reference material (CRM) 7, 87,  
153, 156, 188, 191, 193, 291, 296  
check material homogeneity/stability  
91, 97–99  
chromatographic assays (CC) 142  
classic validation procedure 109, 110  
coefficient of correlation 45, 46  
coefficient of determination 35, 36, 40,  
45–47, 52, 132  
composed standard uncertainty 151

conformity assessment 89, 124, 211, 216,  
217, 227–231, 250, 263, 296, 297,  
301  
constitutional heterogeneity coefficient  
234  
control chart 87, 99–102, 114, 125, 128,  
144, 147, 160, 186, 191–200  
coverage interval 151  
  of given concentration 215  
  of given relative uncertainty 215–216  
  of limits 216–217  
  of origin 211–215

## d

decision limit and detection capability  
  definitions 262–264  
  example of calculation 267–269  
  initial procedure 265  
  modified procedure 265–267  
decision rule, defined 227–229  
decision *versus* uncertainty 221–223  
deuterium effect 8  
direct calibration 27, 28  
direct internal calibration 20–23  
distributional heterogeneity 234  
diverse precision parameters 59–60  
Dumas method 284–287

## e

education about uncertainty 292  
error of measurement 149  
estimators by statisticians 108  
expanded uncertainty 89, 151, 169–171,  
189–191, 206, 229–231, 282, 297  
expected relative uncertainty 260–262  
external calibration (EC) 9–14, 23, 25,  
28, 49, 54, 55, 271, 295  
external reference materials (ERM) 87,  
188

## f

figures-of-merit 108  
fitness-for-purpose 223, 262, 297

framework for decision-making  
  decision *versus* uncertainty 221–223  
  specification limits and reference  
    values 223–226  
fundamental uncertainty 222

## g

generic calibration curve 27  
generic cause to effect diagram 182–184  
generic measurement model 181–182,  
184, 187, 189, 201  
gold standard method 191, 283  
guard band 225, 229–231, 266, 297

## h

harmonization of MU estimation  
  procedures 293–294  
homogeneity check, procedure 236–237  
H-point standard addition method  
  (HPSAM) 17, 272, 283

## i

inconsistencies of validation vocabulary  
107–109  
in-sample calibration (ISC) 10, 11,  
15–17, 26, 272  
interlaboratory comparison data  
200–203  
intermediate precision condition 60, 61,  
65, 70–72, 113, 135, 159, 160, 181,  
182, 187, 229, 237, 251, 253–254,  
264, 275, 290, 298  
intermediate precision variance 61, 69,  
127, 182, 188, 298  
internal calibration 10, 18, 21, 23, 26  
internal reference material (IRM) 8, 87,  
88, 160, 188, 193, 194  
intra-laboratory standard deviation 160  
inverse calibration 25–29, 33, 35, 36, 258  
isotope dilution mass spectrometry  
  (IDMS) 18–20  
isotope inversion 14  
isotopic pattern deconvolution (IPD) 11,  
18–20, 36

**k**

Kjeldahl and Dumas method comparison  
285–287

Kragten iterative algorithm 165–169

**l**

law of propagation of uncertainty

161–164, 181, 184, 187, 199, 201,  
244, 253

least-squares regression method 28–34,  
110

ligand binding assays (LBAs) 142

limit of detection (LOD) 106, 170, 257,  
298

limit of quantification (LOQ)

assessment of 258

definitions of 258

expected relative uncertainty 260–262

linearity, defined 47–48

**m**

matrix-matched external calibration

(MMEC) 10, 14, 55

maximum residue limits (MRL) 206,  
225, 263, 264

measurand (Analyte) 1–9, 154–157

measurement uncertainty (MU) 1, 30,  
272

accuracy, total error, and uncertainty  
171–173

expanded uncertainty 169–170

general procedure 150–151

insight on probability 174–177

Kragten iterative algorithm 165–169

law of propagation of uncertainty  
161–164

measurand 154–157

principle of 149–150

rounding of results 170–171

traceability at the International System  
of units 152–154

uncertainty components 157–158

uncertainty sources 158–161

measurement uncertainty and decision

calibration model 240–243

framework for decision-making

analytical report role 226–227

decision *versus* uncertainty 221–223

specification limits and reference

values 223–226

number of replicates 250–251

replication under intermediate

precision condition 253–254

replication under repeatability

condition 251–253

sample conformity assessment

decision rule 227–229

guard band concept 229–231

sampling uncertainty

example of copper in wheat flour  
237–239

and heterogeneity 232–236

homogeneity check procedure  
236–237

uncertainty of corrected results  
243–249

measurement uncertainty in analytical  
sciences

control charts data 191–200

coverage interval 211–217

interlaboratory comparison data  
200–203

method accuracy profile data 181–191

published procedures 179–181

uncertainty functions 203–211

Metre Convention 149

method accuracy profile (MAP) 7, 26,  
56, 70, 89, 107, 112–122, 156,  
181–191, 241, 243, 261, 281–283,  
294

method comparison using uncertainty

analyte defined by the operating  
procedure 283–284

Kjeldahl and Dumas method  
comparison 285–287

method validation 227

accuracy profile 131–145

method validation (*contd.*)  
 method accuracy profile (MAP)  
 113–122  
 review of validation procedures  
 105–113  
 statistical dispersion intervals  
 122–131  
 metrology 1, 2, 51, 81, 86–89, 108, 149,  
 150, 188, 212, 257, 294, 295  
 misuses of regression 45–48  
 multiple blank standard deviations  
 258–259  
 multiple isotopologue reaction  
 monitoring (MIRM) 21  
 multipoint matrix-matched external  
 calibration (MMEC) 55

**n**

nonlinear calibration curve 41–44, 47  
 normalized response ratio 8  
*n*-way ANOVA 187

**o**

one-factor ANOVA 187  
 one-way fixed effects ANOVA 64  
 one-way random effects ANOVA 64, 236  
 optical density (OD) 25, 43, 161, 274  
 ordinary least-squares (OLS) 29, 30,  
 34–37, 67  
 organization of proficiency testing  
 schemes 90–91

**p**

partial least-squares (PLS) 7, 53, 110, 287  
 performance scores 93–94, 108  
 precision  
 analysis of variance (ANOVA) 64–75  
 ANOVA classic algorithm 72–75  
 balanced and unbalanced experimental  
 design 71–72  
 detect outliers and stragglers 75–78  
 diverse precision parameters 59–60  
 prediction interval 123, 124, 176, 213  
 proficiency testing 89, 160  
 proficiency testing scheme (PTS) 79,  
 89–91, 98, 200, 236

proportion of nonacceptable measures  
 145–147

**q**

quality control (QC) 46, 90, 99, 100, 102,  
 125, 142–144, 147, 160, 191, 193,  
 226, 227, 271, 273, 300  
 quantification  
 authentic *versus* surrogate 3–6  
 calibration 2–3, 9–10  
 direct internal calibration with labeled  
 calibrant 20–23  
 external calibration (EC) 10–14  
 in-sample calibration (ISC) 15–17  
 isotopic pattern deconvolution (IPD)  
 18–20  
 signal pretreatment and normalization  
 6–9

**r**

random effect factor 73, 125, 187  
 reference materials 18, 56, 87–88, 109,  
 134, 153, 156, 188, 193, 194, 199,  
 291, 296, 300  
 reference value  
 of test material 91–93  
 value uncertainty 198–200  
 relative standard deviation (RSD) 20,  
 69–70, 82, 117, 173, 204, 205, 258  
 relative uncertainty 151, 153, 170, 173,  
 176, 190, 199, 208–211, 215–217,  
 224, 230, 231, 238, 239, 242, 245,  
 248, 249, 260–262, 277, 278, 282,  
 285  
 repeatability condition 59, 60, 64, 72, 98,  
 113, 135, 182, 236, 237, 251–253,  
 275–277, 290, 300  
 repeatability variance 61, 62, 69, 72, 136,  
 201, 202, 276, 279  
 reproducibility condition 60, 63, 68, 300  
 reproducibility variance 8, 9, 63, 69, 202  
 Resource H  $\beta$ -expectation tolerance  
 interval (Excel) 117  
 risk analysis 222, 225, 292–293  
 robust estimators 78, 92, 95  
 role of replicates 290

**S**

sample conformity assessment 89, 216, 217, 227–231, 263

sampling uncertainty  
     example of copper in wheat flour 237–239  
     homogeneity check procedure 236–237  
     sampling and heterogeneity 232–236

SARS-CoV-2 virus (COVID-19) 289

scientific uncertainty 222

sensitivity coefficient 164, 168, 182, 201

Shewhart control chart 101, 191–195, 197

signal-to-noise ratio 259, 262, 266

sources of uncertainty in the laboratory  
     machine/equipment 186  
     manpower 185  
     material and handling of items 185  
     measurement and other sources 186–187  
     method 185–186

stable isotope-labeled (SIL)  
     chemicals 8  
     molecule 13

standard addition method (SAM) 15, 48, 272  
     estimation from method accuracy profile 281–283  
     without replication 273–279  
     with replication 279–281  
     and surrogate samples 88–89

standard deviation of grand mean 72

standard uncertainty 89, 92, 151, 152, 158, 159, 161–163, 165–167, 169, 170, 172, 173, 183, 184, 187–189, 191, 195, 199–202, 206, 208–211, 215, 238, 244, 245, 251–253, 269, 276, 296, 300, 301

statistical dispersion intervals 112–114, 122–131, 141, 143, 195–198, 212, 295, 296, 301, 302

statistical process control (SPC) 99, 123, 193

straight-line computation 28–31

surrogate calibrant in authentic matrix 13–14

surrogate standard 3–5, 13, 14, 20, 21, 56, 296

**t**

10-step MAP procedure 114, 295

theophylline 26–28, 36, 39, 40, 116–121, 127, 129–133, 136, 138–141, 143–146, 152, 153, 155, 190–192, 208–210, 215, 241–243, 252, 253, 261

total analytical error (TAE) 111, 112, 149, 150, 173, 190

toxicological reference values (TRV) 225

traceability  
     International System of units 152–154  
     to International Units 290–291

transmittance 25

trueness  
     algorithm A 94–97  
     bias and recovery yield 82–83  
     check material homogeneity/stability 97–99  
     control charts 99–102  
     evolution of 83–84  
     organization of proficiency testing schemes 90–91  
     performance score 93–94  
     primary operating procedures 86  
     reference materials 87–89  
     reference value of 91–93  
     specificity and sources of bias 84–86

**U**

uncertainty components 56, 157–158, 160, 164

uncertainty functions 57, 113, 168, 176, 177, 182, 203–211, 215–217, 224, 229, 231, 236, 238, 242, 243, 245, 248, 249, 251–253, 260, 261, 269, 285, 286

uncertainty sources 53, 56, 158–161, 183

upper limit of quantification (ULOQ)  
     12, 120, 260